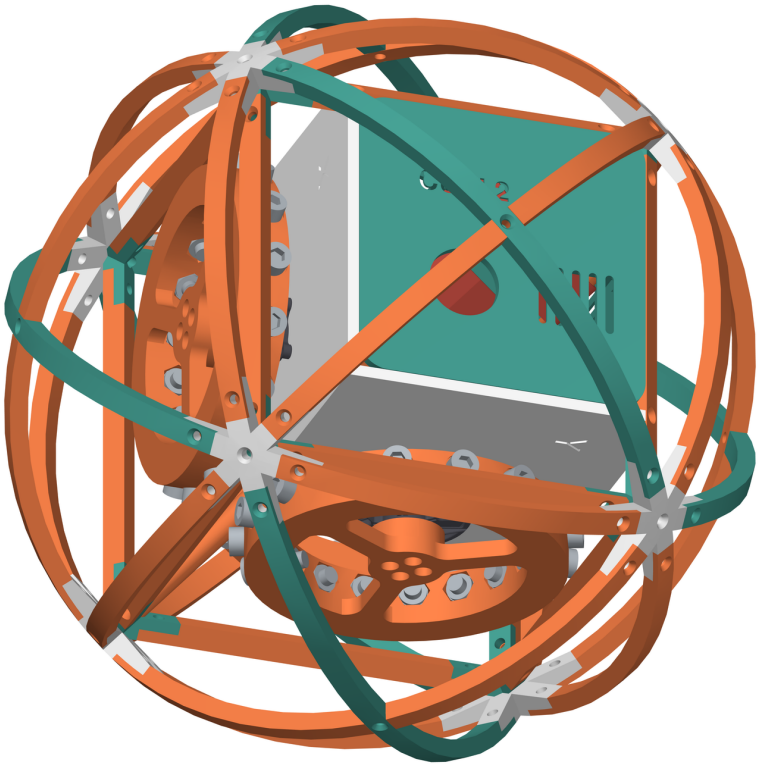
Assembly

Building the cube, step by step



Designed by Athanasia Nikolova

v1.1 · October 2026

All data come from tests and many real builds, not only from theory.

Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

✋ Safety first

- Work with your teacher or mentor in the room.
- NO battery until the step that says so - near the end of this guide.
- The wheels are heavy. Once the cube runs, fingers away from them.
- Screws go into plastic: turn until it stops, then no more. Too hard and the thread is gone.
- Small screws and magnets away from small children.

Before you start

- Inserts guide: every insert is in its part.
- Magnets guide: the 3 magnet carriers are on the motors, the cover magnets are in.
- Boards guide: the 3 boards are built and checked.
- All printed parts are ready.

The order

1 Motors and wheels

2 Battery holder and IMU

3 Wiring - with the Wiring guide open next to this one

4 The cage

5 Battery and covers

6 The cube alone - 8 corner tips (try it out)

7 The sphere

You need

- Hex keys 1.5, 2, 2.5, 4 and 5 mm - a ball-end 2.5 mm helps at the corners
- A 10 mm spanner (the wheel nuts)
- A tray for the screws - one cup for each kind
- A soft cloth to work on, 4 toothpicks
- The screws: sort them first (next pages)



hex keys



spanner



soft cloth



flat table

How to read the pictures

- Colour = what you fit now. Grey = already there.
- The part floats where it comes from. The arrow shows the way in.
- Under each step: what to watch, then small pictures for the tool and the way to do it.
- Each screw is drawn at its REAL size. Print this guide at 100 % and lay the screw on its picture: same length, same head - right screw.
- ×3 at the top right: do the same 3 times (faces X, Y, Z).

The small pictures


Start every screw by hand, then the key.



4 screws: tighten across - 1, 2, 3, 4 - a little each time.


Countersunk head: level with the surface, never deeper.


Never force a part. If it does not go, look why.


Look closely before the next step.

Sort the screws

Real size. Lay each one on its picture.

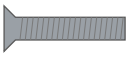
54 ×  **M3 x 6**
key 2 mm · DIN 7991

28 ×  **M3 x 8**
key 2 mm · DIN 7991

2 ×  **M3 x 10**
key 2 mm · DIN 7991

3 ×  **M3 x 12**
key 2 mm · DIN 7991

3 ×  **M3 x 14**
key 2 mm · DIN 7991

6 ×  **M3 x 16**
key 2 mm · DIN 7991

12 ×  **M2 x 5**
key 1.5 mm · DIN 912

12 ×  **M2.5 x 5**
key 2 mm · DIN 912

2 ×  **M2.5 x 6**
key 2 mm · DIN 912

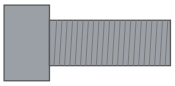
4 ×  **M3 x 6**
key 2.5 mm · DIN 912

18 ×  **M3 x 8**
key 2.5 mm · DIN 912

12 ×  **M3 x 12**
key 2.5 mm · DIN 912

2 ×  **M3 x 20**
key 2.5 mm · DIN 912

2 ×  **M5 x 6**
key 4 mm · DIN 912

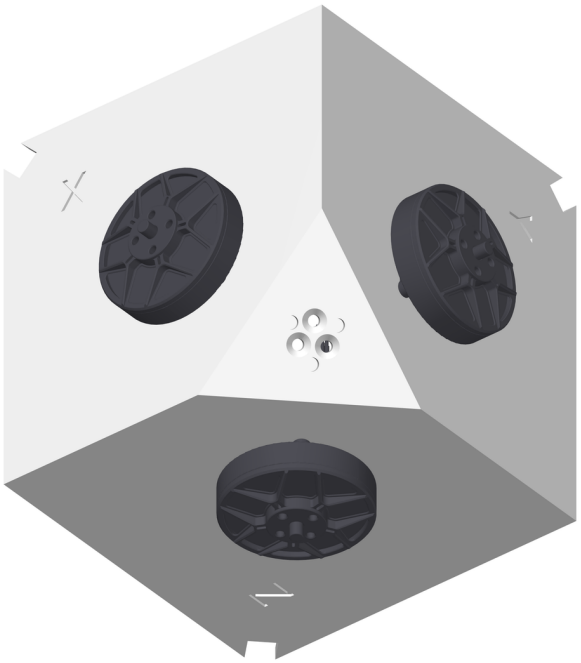
48 ×  **M6 x 16**
key 5 mm · DIN 912

8 ×  **M3 x 12**
key 1.5 mm · DIN 914

48 ×  **M6 nut**
spanner 10 mm · DIN 985

1 · Motors and wheels

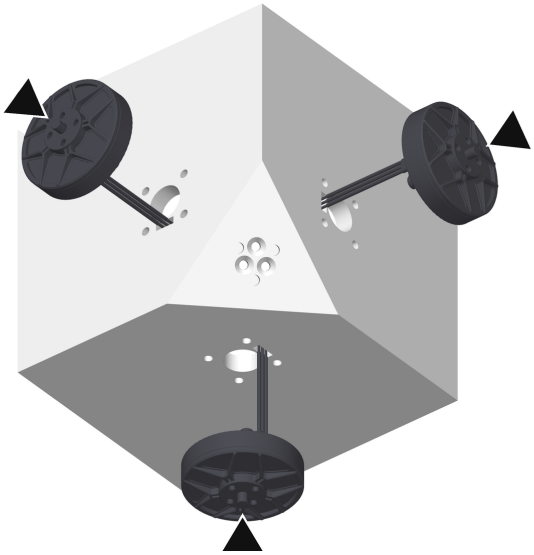
The 3 motors on the 3 closed faces, a wheel on each.



- Work on a soft cloth: the box does not slide or scratch.
- The magnet carriers are already on the motors (Magnets guide).
- Do one face completely, then the next - X, Y, Z.

1 Motor onto the face

×3 · faces X, Y, Z



Motor on the outside of the face, its 3 leads in through the big hole.

- Turn the motor until its 4 holes sit over the 4 holes of the wall.
- The 4 screws go in from INSIDE the box into the motor.
- Pull the leads through gently - never pinch them between motor and wall.



by hand first



crosswise

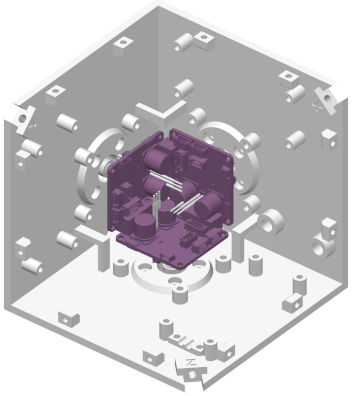


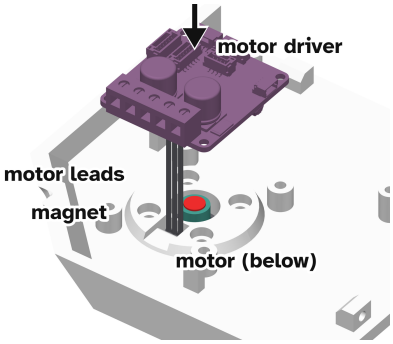
key 2.5 mm

4 ×  **M3 x 8**
key 2.5 mm

2 Motor driver

×3





close-up

Inside the box, over the motor's magnet: 4 screws.

- Only now: the motor is on its face and its magnet is glued on (Magnets guide). Its 3 leads stand up beside the board.
- The small sensor in the middle of the board sits right over the magnet.
- Look from the side: a small gap between magnet and board. Nothing touches.
- The 3 motor leads into its U, V, W terminals - in any order: the driver finds the order itself when it calibrates (software).



the gap



no force

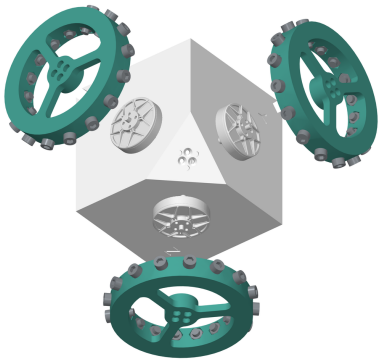


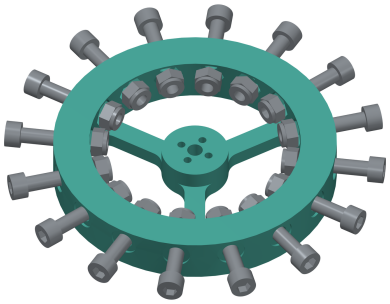
key 2 mm

4 ×  **M2.5 x 5**
key 2 mm

3 Rim masses into the wheel

×3 · 3 wheels





close-up

16 screws through the rim, a nut on each, from the inside.

- All 16 the same way: head outside, nut inside - the wheel stays balanced.
- The nyloc nut turns hard - that is normal. Hold the head with the key, turn the nut with the spanner.
- Tight: the masses spin fast, they must not move.



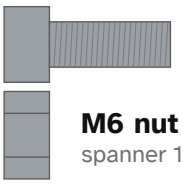
key 5 mm



spanner 10



on the table

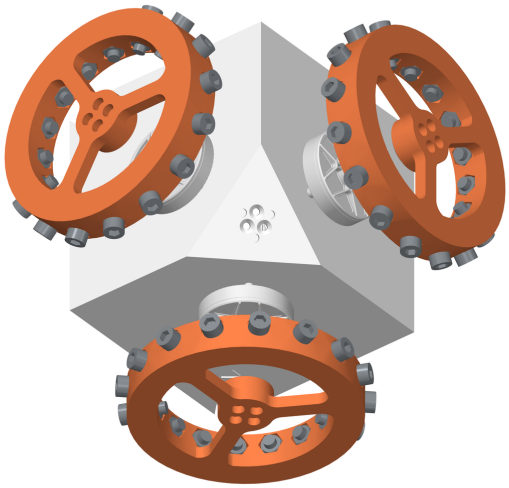


16 × **M6 x 16**
key 5 mm

16 × **M6 nut**
spanner 10 mm

4 Wheel onto the motor

×3



The thin metal ring on the motor, then the wheel, 4 screws.

- The ring sits centred on the turning part of the motor.
- The wheel's 4 holes over the 4 holes under the ring.
- Turn the wheel by hand: round and free, it never touches the box.



by hand first



crosswise

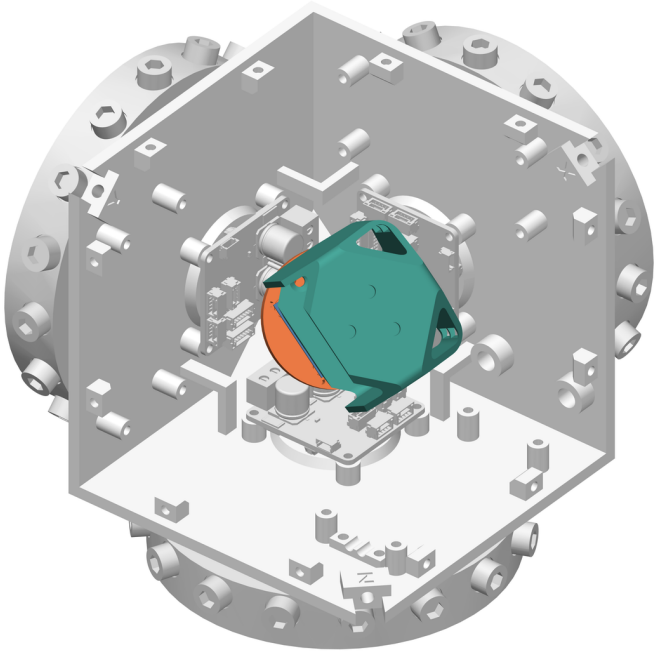


turn and look

4 ×  **M3 x 12**
key 2.5 mm

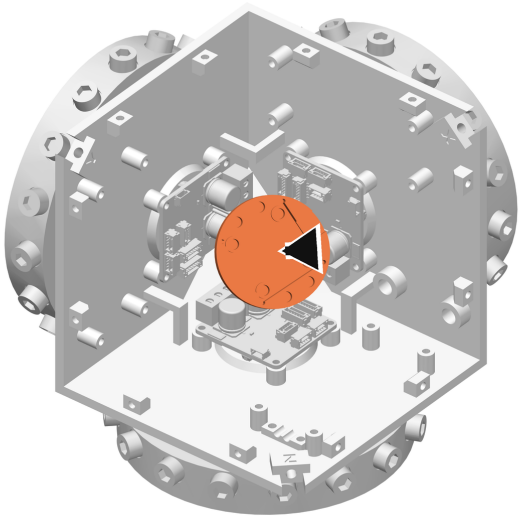
2 · Battery holder and IMU

Along the long diagonal, from corner 0 - the corner where the 3 motor faces meet.



- The IMU tells the cube how it is tilted. Its position matters: read its step twice.
- The straps go through the basin BEFORE the basin goes in.

1 Battery tube to corner 0



Tube in from inside, its foot on corner 0. 3 screws from OUTSIDE.

- Turn the tube until its 3 holes sit over the 3 holes in the corner.
- Countersunk heads: level with the outside, not deeper.



by hand first



flush

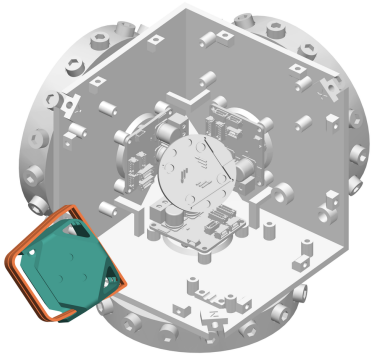


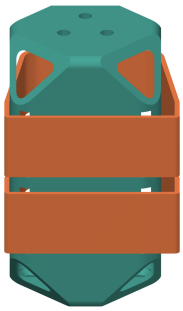
key 2 mm

3 × 

M3 x 12
key 2 mm

2 Straps through the basin





close-up

The 2 hook-and-loop straps through the slots in the back of the basin.

- Now, while the basin is in your hands.
- Each strap through its own pair of slots.
- Leave them open: they close round the battery at the end.

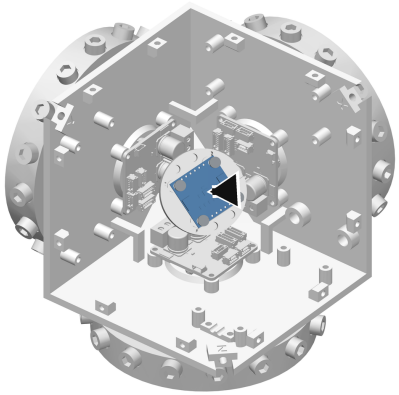


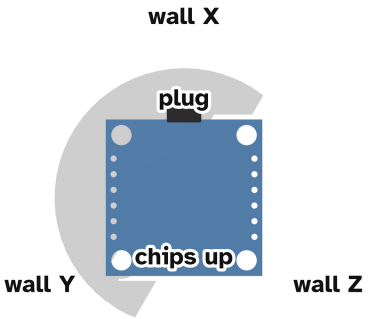
2 straps



on the table

3 IMU into the table





close-up

The IMU into the pocket of the table, 4 screws M3x6.

- The IMU's parts (its chips) face UP - towards the basin and the middle of the cube. Its 4-pin plug points to the wall with the letter X, into the table's slot.
- The board sits on the pocket's 4 small bosses. The pocket's corners are rounded out: it goes in by hand.
- Tighten across, a little each time: the board must not bend.



plug to X



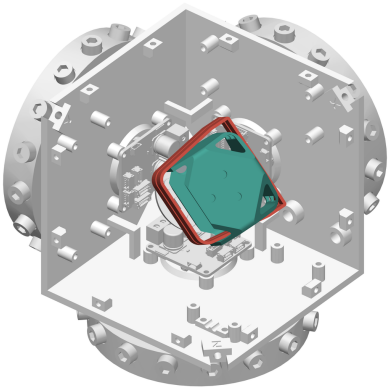
no force

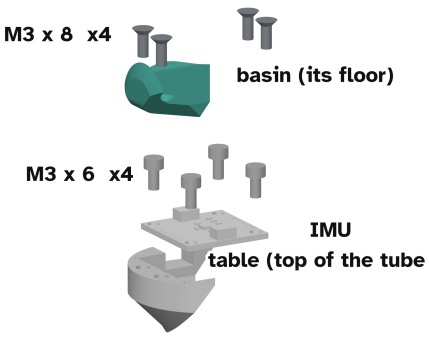


crosswise

4 ×  **M3 x 6**
key 2.5 mm

4 Basin onto the table





close-up

The basin on the table, 4 countersunk screws M3x8 from inside the basin.

- The basin's floor lies flat on the table: nothing touches the IMU.
- The straps are already through the basin (step 2).
- Heads level with the floor - the battery lies on it. Tighten across.



crosswise

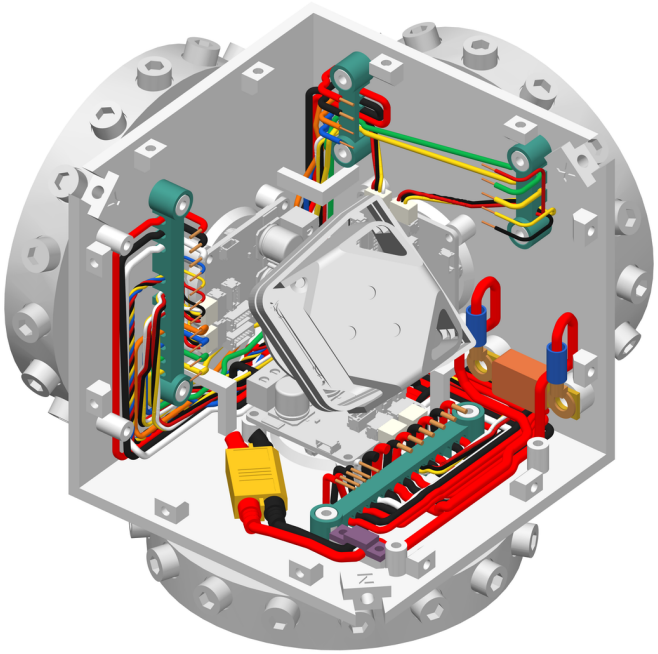

flush


key 2 mm

4 ×  **M3 x 8**
key 2 mm

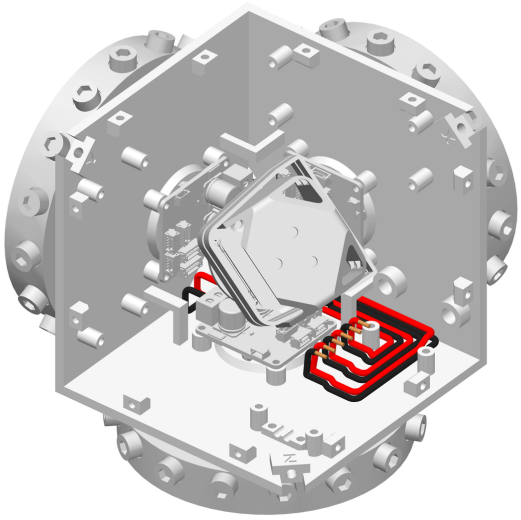
3 · Wiring

All the wires, in 5 stages. Each wire, step by step, is in the Wiring guide: work with it open next to this one.



- The Wiring guide has its own safety page, cut lists and checks.
- Still NO battery.
- Lowest wires first: each wire lies on the ones under it.

1 Motor power



6 thick wires from the 3 motor drivers to board Z.

- They lie flat on the floor, side by side, red and black in turn.
- Each wire where its arch of the cable clamp will be. The clamps and the boards go on later, over the wires.
- Wiring guide: chapter 1.

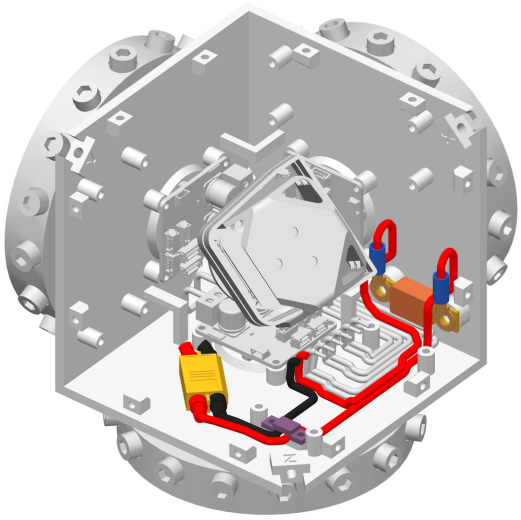


strip 7 mm



one per arch

2 Battery supply



Fuse, battery connector and cable clamp.

- The fuse's 2 lugs on their M5 screws, the clamp over the thick wires.
- The battery's own half of the connector: teacher only.
- Wiring guide: chapter 2.



fuse



solder

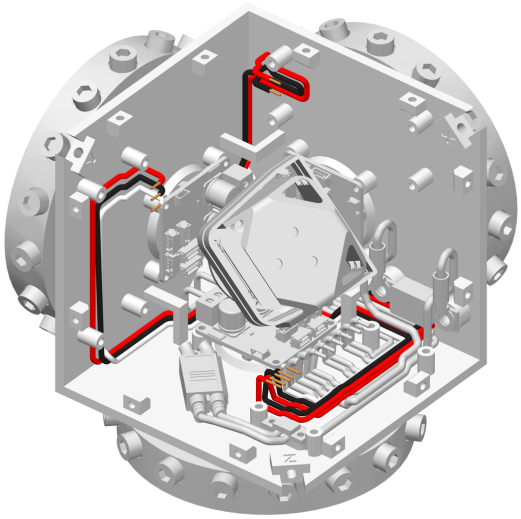


heat-shrink

2 ×  **M5 x 6**
key 4 mm

2 ×  **M2.5 x 6**
key 2 mm

3 Board power



The 5 V wires from board Z to boards Y and X, and SENSE.

- They lie on top of the motor wires.
- Wiring guide: chapter 3.

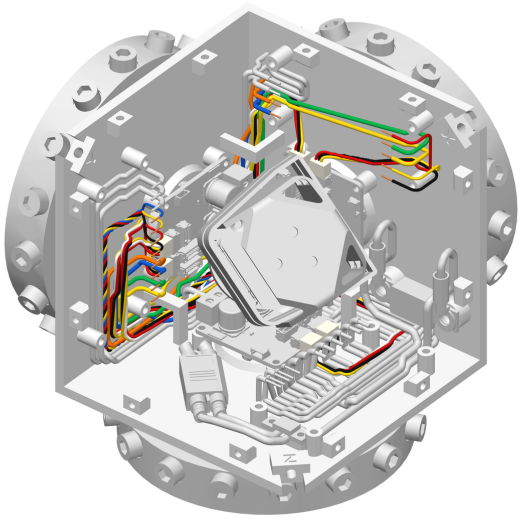


strip 7 mm



one per arch

4 Signals



The links, the CAN cables and the I2C cable.

- Thin wires: never pull them tight round a corner.
- Wiring guide: chapter 4.



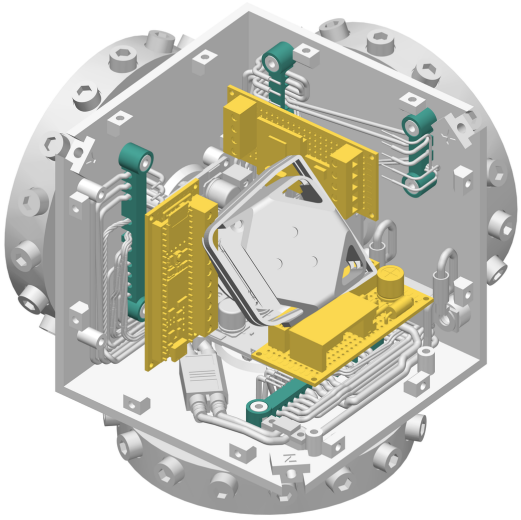
signals



no pulling

5 Boards on

×3 · 4 each



The 4 clamps over the wires, a spacer on each ring, the 3 boards on their standoffs.

- 4 screws per board. Board Z first, then Y, then X.
- Each wire end now stands in front of its terminal. Do not connect yet.



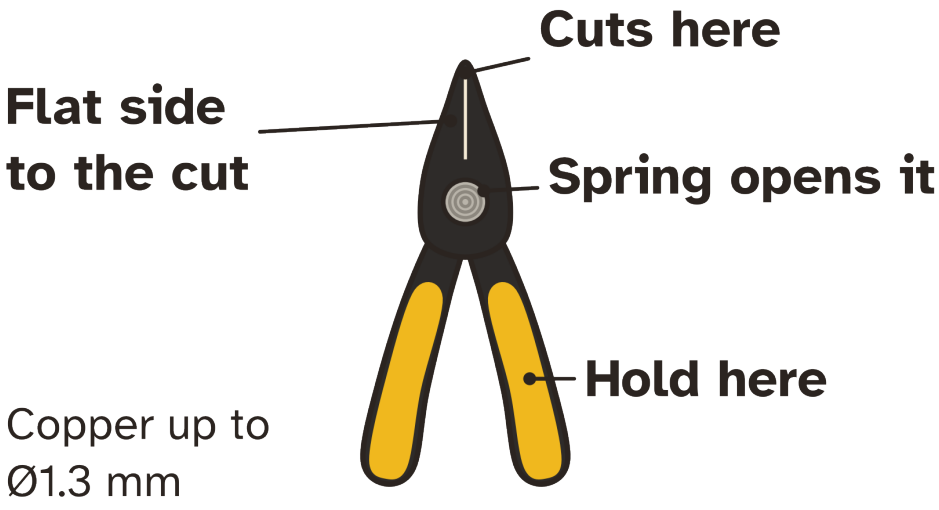
key 2 mm



by hand first

12 ×  **M2 x 5**
key 1.5 mm

6 Mark and cut



Each wire end: a mark with a permanent marker 7 mm past its terminal's opening, cut there.

- Hold the wire against its terminal to find the place. One wire at a time.
- The side cutter: its flat side towards the cut.
- The pole name (z1, y3 …) is on the wire's name flag and on the terminal map.

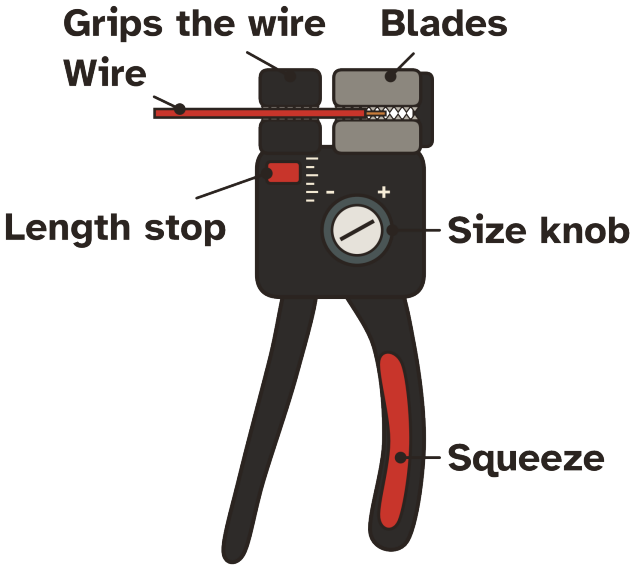


marker



side cutter

7 Strip 7 mm



The wire stripper set to the wire size, the wire in up to 7 mm, squeeze.

- 7 mm of bare copper - no more.
- Stranded wire: twist the strands tight, no solder on them. Solid wire: straight as it is.

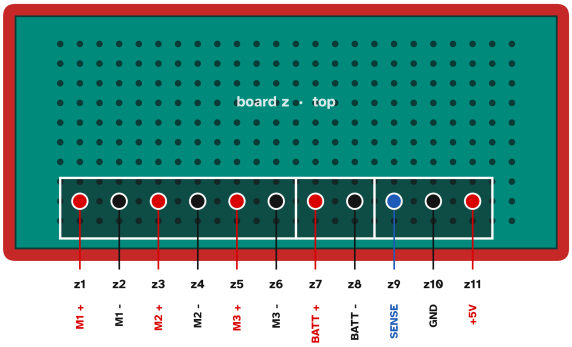


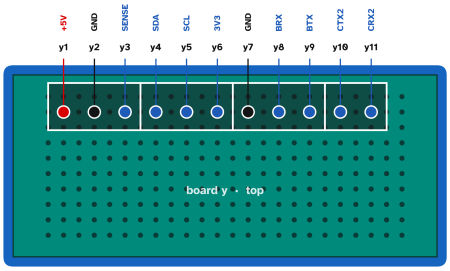
7 mm



no nicks

8 Into its terminal





close-up

Each end into the pole of its name - the maps: board Z here, Y beside it, X below.

- All the copper in, none outside. Tighten with the small screwdriver.
- Pull gently: it must not move. Then the plugs: CAN into the drivers, I2C into the IMU.
- Final check with the multimeter - in the Wiring guide. Still NO battery.



terminals

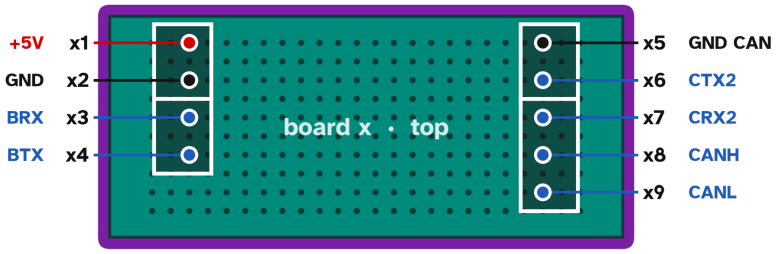


plugs



check

9 Board X terminals



Board X: the two blocks, one on each side.

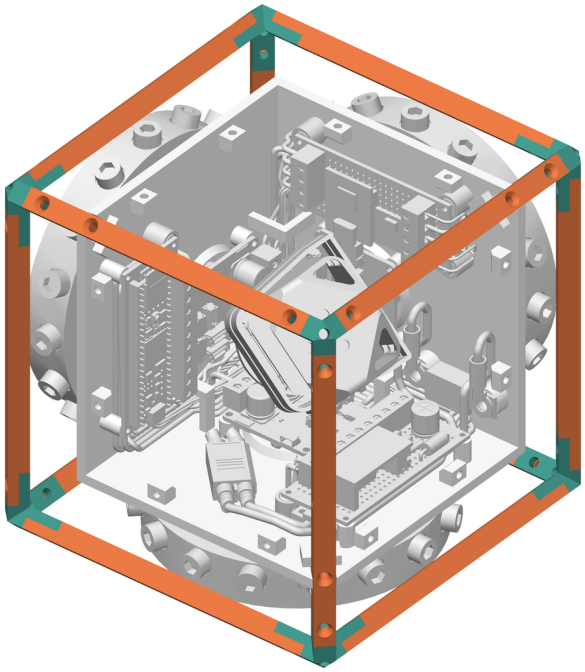
- x1 - x4 on the west block, x5 - x9 on the east block.



terminals

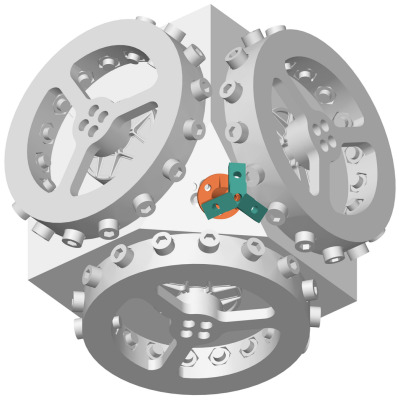
4 · The cage

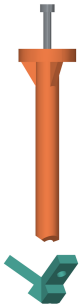
8 corners, 12 beams. It holds the box at 2 corners and carries the sphere.



- Corner 0 and the far corner (1,1,1) are the 2 corners with 2 inserts.
- Countersunk screws: level with the surface, never deeper.
- Every cage screw snug first; tighten them all when the frame is closed.

1 Box standoff into corner 0





close-up

On the table: the long screw down through the standoff into the corner.

- Take a corner with an insert INSIDE (one of the 2 corners with 2 inserts).
- The head goes right down into its round pocket - level with the end of the standoff.
- Tight: the standoff must not turn.



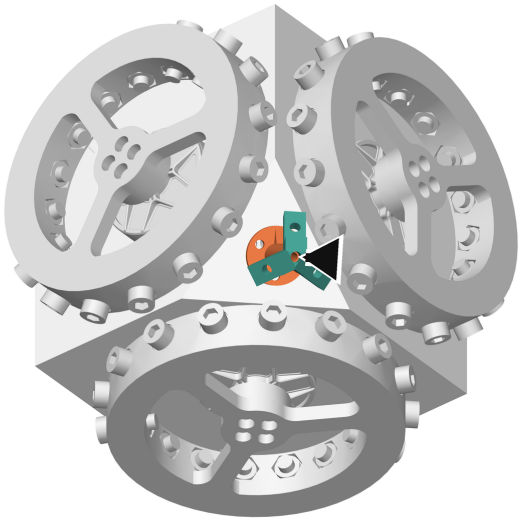
on the table



long key 2.5

1 ×  **M3 x 20**
key 2.5 mm

2 Corner 0 onto the box



The round flange flat on corner 0 of the box. 3 screws through the flange.

- Nothing may lie under the flange.
- Its 3 screws sit between the 3 heads of the tube screws.
- A ball-end key reaches past the cage corner.



by hand first



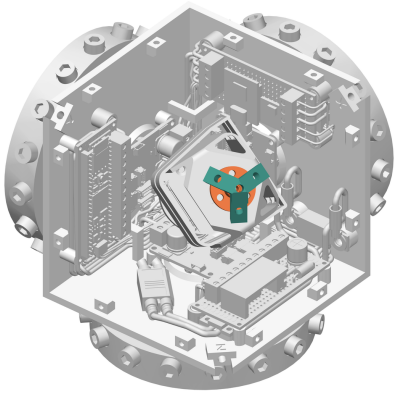
across

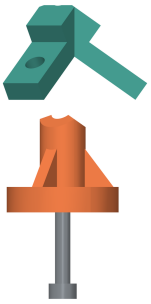


ball-end 2.5

3 ×  **M3 x 8**
key 2.5 mm

3 Hub standoff into the far corner





close-up

On the table: the long screw up through the hub standoff into the 2nd corner with 2 inserts.

- The top of the standoff has the shape of the corner: turn it until it sits.
- Head level with the flange.



on the table



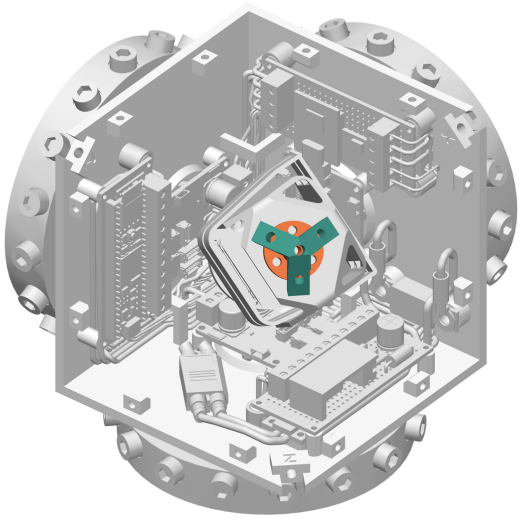
it sits



long key 2.5

1 ×  **M3 x 20**
key 2.5 mm

4 Far corner onto the basin



Through an open face: the flange flat on top of the basin, 3 screws.

- No covers yet - work through the open faces.
- Do not pull on the basin: the IMU stack holds it.



by hand first



no pulling

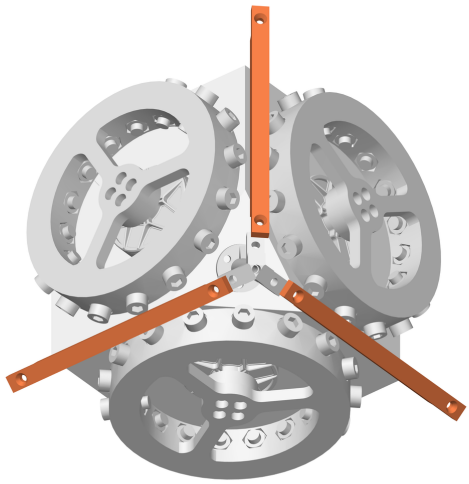


key 2.5 mm

3 × 

M3 x 8
key 2.5 mm

5 3 beams at corner 0



Each beam lies on the corner from the outside. 1 screw each.

- Plain beams here (no magnets).
- Snug only - tighten when the frame is closed.



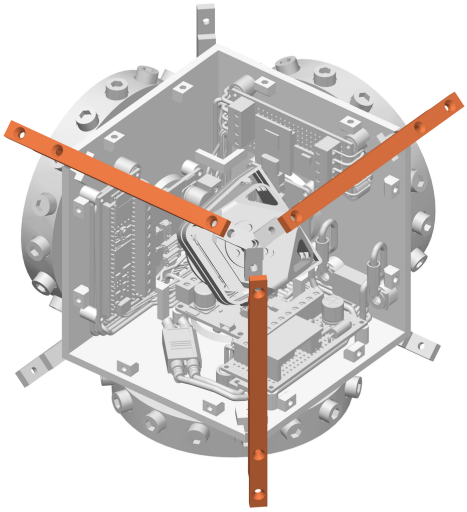
flush



snug

3 ×  **M3 x 8**
key 2 mm

6 3 tie beams at the far corner



The 3 beams with magnets: 1 screw into the corner, the long one into the box.

- Their magnets face the covers.
- The long screw (M3 x 14) goes through the beam into the box wall.



magnets in

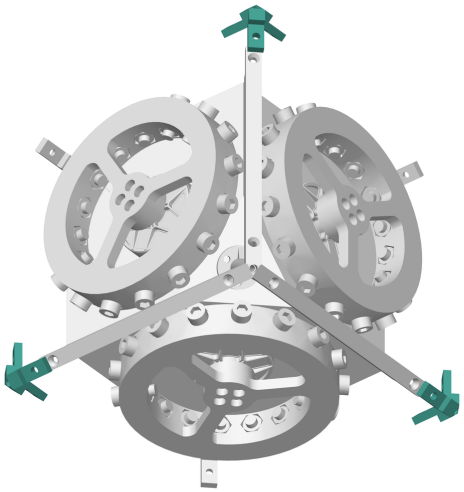


flush

3 ×  **M3 x 8**
key 2 mm

3 ×  **M3 x 14**
key 2 mm

7 3 corners on the corner-0 beams



Each corner closes on the free end of a beam. 1 screw each.

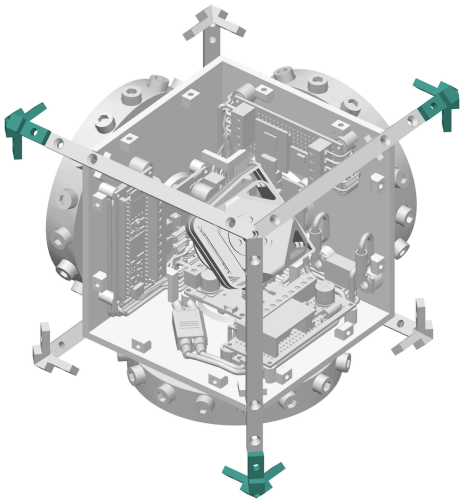
- Snug only.



snug

3 ×  **M3 x 8**
key 2 mm

8 3 corners on the tie beams



Each corner closes on the free end of a tie beam. 1 screw each.

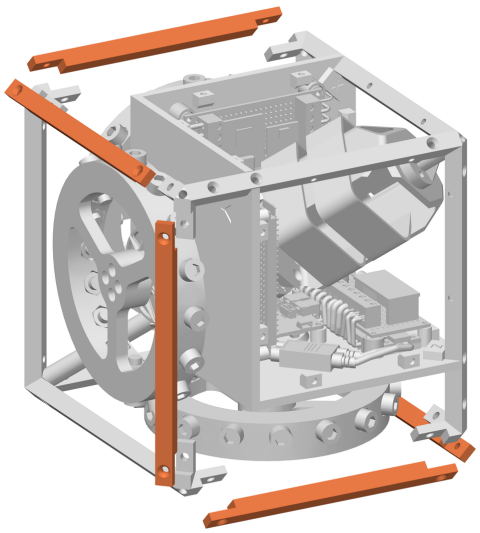
- Snug only.



snug

3 ×  **M3 x 8**
key 2 mm

9 The last 6 beams



Each between two corners, 2 screws.

- Now tighten every cage screw.
- Shake gently: nothing moves.



flush

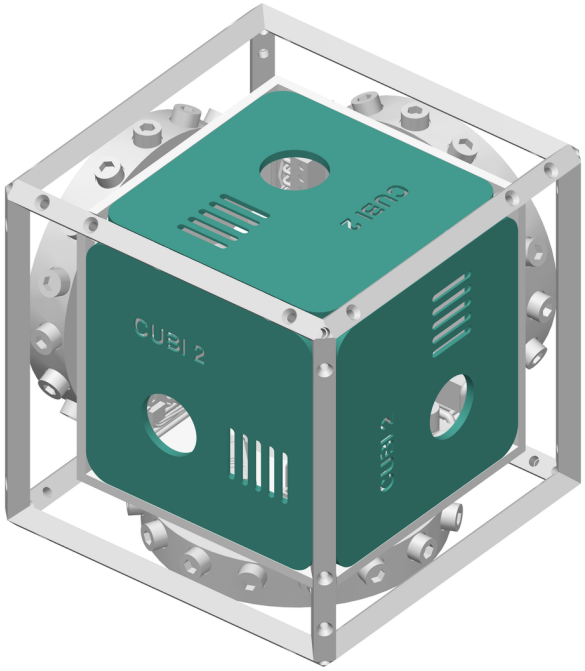


check

12 ×  **M3 x 8**
key 2 mm

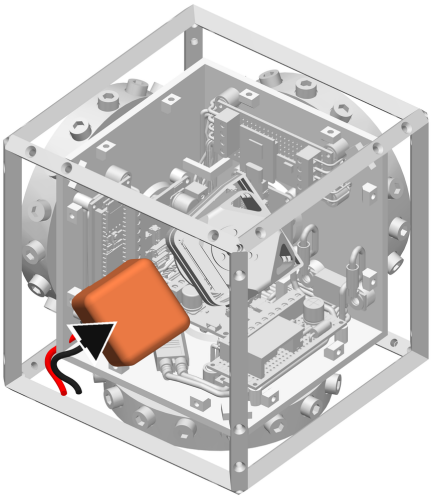
5 · Battery and covers

Only with your teacher.



- Never force the battery. Never let its two leads touch.
- A battery that is puffy, hot or damaged is not used.

1 Battery in



Slide the battery in from the X side, across the diagonal, under the open straps.

- Its connector end first, towards corner 0.
- Pull each strap tight round the battery and close it: the straps hold it in the basin.
- Plug it in only together with your teacher: the first power-up (Wiring guide, final check). The connector fits only one way: red + to +. Never force it.



straps tight

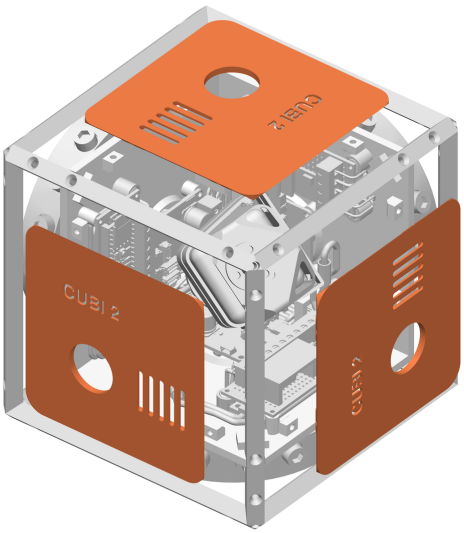


no force



with teacher

2 Covers on



Each cover on its own face. The magnets pull it in.

- Its letter (X, Y, Z) inside matches the box.
- Look first: no wire on the edge - the cover would pinch it.
- To take a cover off: pull at its edge.



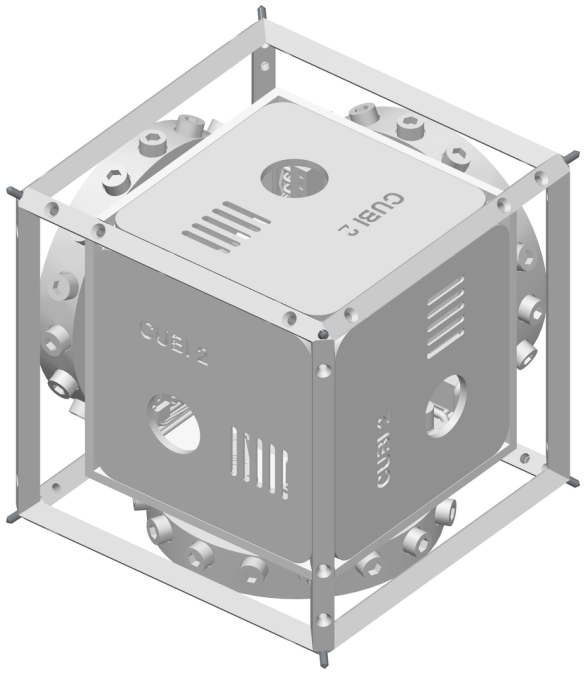
no wire



no force

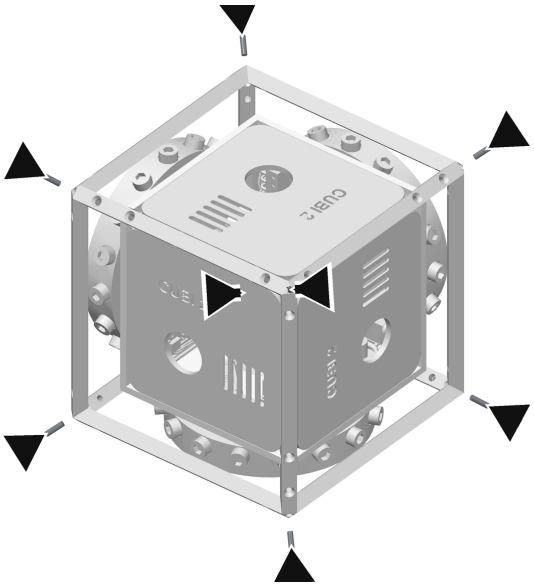
6 · The cube alone

A stage to try it out: the bare cube balances on a corner. 8 corner tips, no sphere yet.



- The tips go into the same inserts as the navels later.
- Try it on a soft mat: the cube can fall over.

1 Tips on the 8 corners



One grub screw with a cone point into the tip insert of each corner.

- The point sticks out 6.2 mm from the cut face - measure with the depth gauge of the caliper. Then the point is exactly where the corner would be: the cube still sits flat on a face.
- Less: it does not reach the ground. More: the cube rocks on 4 tips.
- Before the sphere (chapter 7): all 8 tips come out again.



key 1.5 mm



6.2 mm out

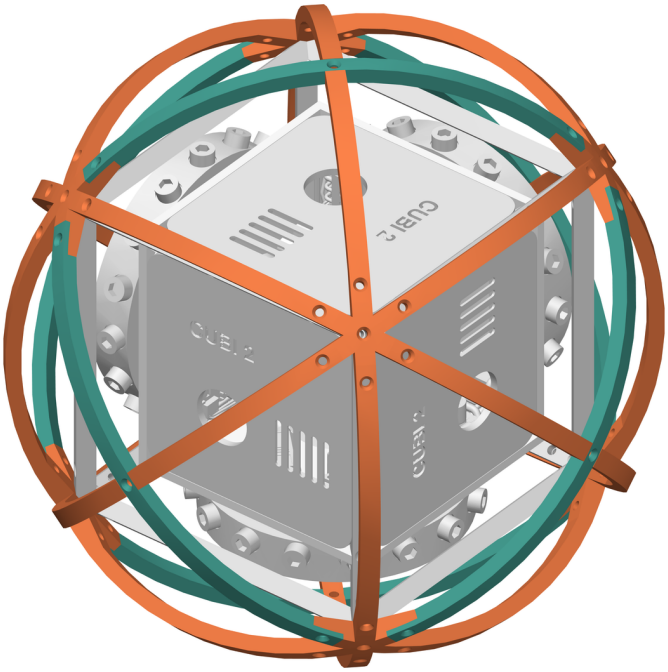


soft mat

8 ×  **M3 x 12**
key 1.5 mm

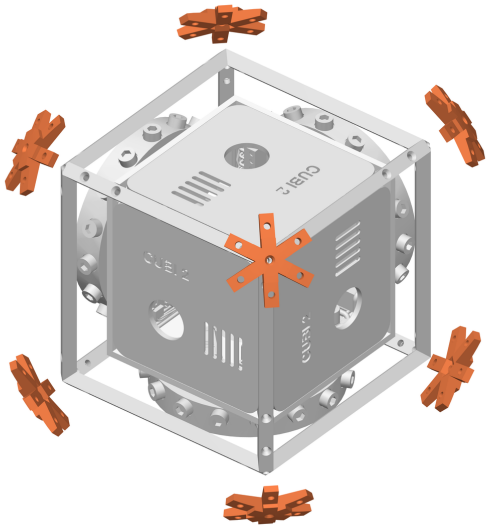
7 · The sphere

8 navels and 24 arcs make the ball the cube rolls in.



- Lay the cube on a soft cloth: the arcs scratch.
- The order: navels, edge arcs, lower diagonal arcs, upper diagonal arcs.
- All screws snug first; tighten them all when the ball is closed - it comes out round.

1 Navels on the 8 corners



One screw in the middle of each navel, into the corner tip.

- First the 8 corner tips out (chapter 6): the navels go into the same inserts.
- Corner 0 and the far corner take the SHORT screw (M3 x 10): their inside is taken.
- Turn each navel so its arms point along the edges and the face diagonals.



soft cloth

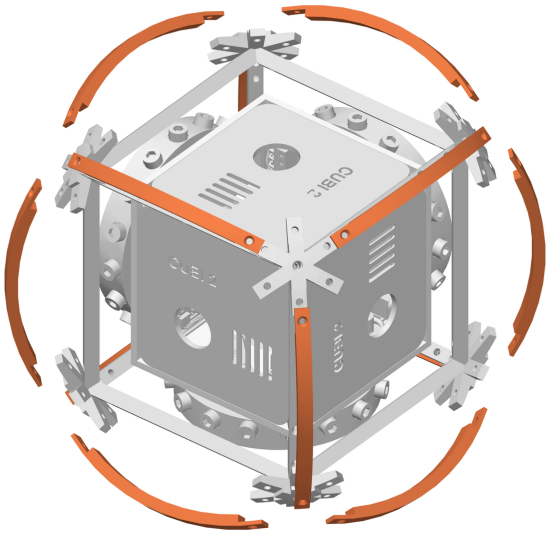


flush

6 ×  **M3 x 16**
key 2 mm

2 ×  **M3 x 10**
key 2 mm

2 Edge arcs



12 arcs, one over each edge. 1 screw at each end.

- Each end into a navel arm.
- Snug only.



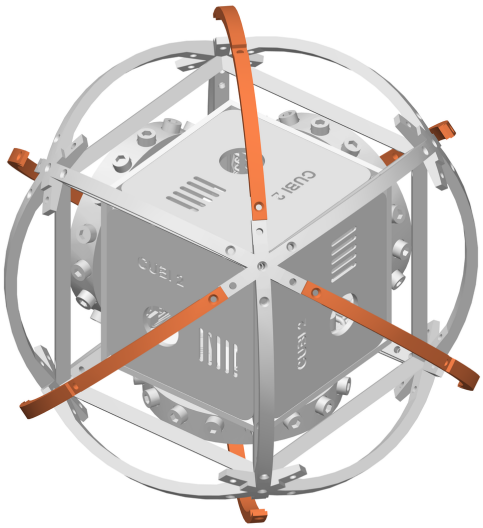
snug



flush

24 ×  **M3 x 6**
key 2 mm

3 Diagonal arcs - the lower ones



On each face, one diagonal arc first: the LOWER one.

- The lower arc has an insert in the middle.
- 1 screw at each end, snug.



insert up



snug

12 ×  **M3 x 6**
key 2 mm

4 Diagonal arcs - the upper ones



The second arc crosses over the first. 1 screw at each end, 1 in the middle.

- The middle screw goes into the lower arc's insert.
- Now tighten every sphere screw. Roll the ball: it is round.



all tight


round?

12 × **M3 x 6**
key 2 mm

6 × **M3 x 6**
key 2 mm

✅ Final check

- Every screw: tight, no more.
- Turn each wheel by hand: free, no rubbing, no click.
- Shake the cube gently: nothing rattles inside.
- Roll it on the floor: the sphere is round, no arc sticks out.
- Spare screws left over? Find where they belong before the first run.